

tenets of utility theory. In particular, people underweight outcomes that are merely probable in comparison with outcomes that are obtained with certainty. This tendency, called the certainty effect, contributes to risk aversion in choices involving sure gains and to risk seeking in choices involving sure losses. In addition, people generally discard components that are shared by all prospects under consideration. This tendency, called the isolation effect, leads to inconsistent preferences when the same choice is presented in different forms. An alternative theory of choice is developed, in which value is assigned to gains and losses rather than to final assets and in which probabilities are replaced by decision weights. The value function is normally concave for gains, commonly convex for losses, and is generally steeper for losses than for gains. Decision weights are generally lower than the corresponding probabilities, except in the range of low probabilities. Overweighting of low probabilities may contribute to the attractiveness of both insurance and gambling.¹³

Prospect theory, in essence, describes how individuals evaluate gains and losses. The theory names two specific thought processes: editing and evaluation. During the editing state, alternatives are ranked according to a basic “rule of thumb” (heuristic), which contrasts with the elaborate algorithm in the previous section. Then, during the evaluation phase, some reference point that provides a relative basis for appraising gains and losses is designated. A value function, passing through this reference point and assigning a “value” to each positive or negative outcome, is S shaped and asymmetrical in order to reflect loss aversion (i.e., the tendency to feel the impact of losses more than gains). This can also be thought of as risk seeking in domain losses (the reflection effect). Figure 2.4 depicts a value function, as typically diagrammed in prospect theory.

It is important to note that prospect theory also observes how people mentally “frame” predicted outcomes, often in very subjective terms; this accordingly affects expected utility. An exemplary instance of framing is given by the experimental data cited in the 1979 article by Kahneman and Tversky, where they reported that they presented groups of subjects with a number of problems.¹⁴ One group was presented with this problem:

1. In addition to whatever you own, you have been given \$1,000. You are now asked to choose between:
 - A. A sure gain of \$500.
 - B. A 50 percent chance to gain \$1,000 and a 50 percent chance to gain nothing.